CS4740/CS5740

Introduction to Natural Language Processing

Name:

Netid:

Instructions (****READ CAREFULLY****):

- IF YOU HAVE ANY QUESTIONS ABOUT A QUESTION ON THE TEST, PLEASE JUST INDICATE THIS IN YOUR ANSWER. State any assumptions that you need to make in order to answer the question and then answer the question as best as you can. Only raise your hand to ask us a question if you believe that there is an error in the test.
- Use a **pen or dark-leaded pencil** so that your scanned answer sheets are readable.
- You have **1.5 hours** to complete this exam. The exam is a closed-book / closed-notes / no phones / no cheat sheet / no laptop exam.
- It is **entirely your responsibility** to ensure that your test is **placed directly into the exam collection box** at the front of the classroom at the end of the examination time.

1 N-gram Language models (16 points)

1. (4 points) In one sentence (i.e., your answer should be just one sentence), what is the purpose of smoothing in n-gram language modeling?

2. (4 points) In one sentence, what is the purpose of replacing rare words in the training corpus with an unknown word identifier?

3. The parameters of a 3-gram language model refer to unsmoothed bigram and trigram counts: count(w_1, w_2, w_3) and count(w_1, w_2) for all word tuples in a corpus. Suppose we have access to these parameters but the actual corpus C is hidden. **Is each of the following statements true or false? Briefly explain each answer.**

   (a) (4 points) We can apply add-k Laplace smoothing to the 3-gram language model using this count information.

(b) (4 points) We can determine the parameters of a 2-gram language model for C from this information, i.e. $\text{count}(w_1, w_2)$ and $\text{count}(w_1)$.

2 Text Classification (24 points)

1. (3 points) Consider the task of classifying a given text passage as either Shakespearean or not Shakespearean. Design 3 features that will be useful for this binary classification task. Clearly define how a numerical value for each feature will be extracted from a given text.

2. (3 points) Write the equation for the logistic function $\sigma(z)$.

3. The rough pseudocode for **binary logistic regression** is given below. Assume $\mathbf{w} = [w_1, w_2...w_k]$ is the weight vector and $\mathbf{f}^j = [f_1^j, f_2^j...f_k^j]$ is the feature vector for datapoint j. Everywhere, the **superscript** denotes the index of the datapoint and the **subscript**

denotes the index of the weight or feature. We also assume a training dataset $\mathcal{D} = \{(\mathbf{f}^1, y^1), (\mathbf{f}^2, y^2)...(\mathbf{f}^N, y^N)\}$, where y^j is the output label.

Algorithm 1 TRAINING BINARY LOGISTIC REGRESSION

1. Initialize $\mathbf{w} =< w_1, w_2, w_3 >$

for j in len(training_data):

   2. Compute Loss for datapoint j, namely L^j
   3. Compute partial derivative $\frac{\partial L^j}{\partial w_i}$

      Hint: for binary logistic regression, $\frac{\partial L^j}{\partial w_i} = f_i^j \left[\sigma\left(\sum_i w_i f_i^j\right) - y^j\right]$
   4. Update each w_i in $\mathbf{w}$

5. Return $\mathbf{w}$

(a) (4 points) What is the equation for the loss for datapoint j, i.e. $L^{(j)}$.

(b) (4 points) What is the update equation in step 4? Use α to denote the the learning rate.

(c) (10 points) Suppose you have the following training data for binary classification, written in terms of features and labels:

$\mathbf{f}^1 =< 1, 0, -1 >, \ \ y^1 = 1$
$\mathbf{f}^2 =< 1, 1, 1 >, \ \ y^2 = 0$

Run binary logistic regression on this data using stochastic gradient descent. You need to directly follow the pseudocode above. Assume **weights** w_1, w_2, w_3 are initialized to $1, 1, 1$ and the **learning rate** is 1. Iterate through the data in the order 1-2 and report the weights w_1, w_2, w_3 after each step. This means you will report 2 values of weight vectors. Show your work.

To simplify calculations, estimate the sigmoid σ as follows:

$$\sigma(z) = \begin{cases} 1/2, & \text{if } z = 0 \\ 1, & \text{if } z > 0 \\ 0, & \text{if } z < 0 \end{cases}$$

3 Word Embeddings (10 points)

1. (4 points) What are two important advantages of dense embeddings (e.g. word2vec) over sparse embeddings (e.g. word co-occurence or tf-idf)?

2. (6 points) You are given the following data (Table 1) to train a **skip-gram model** with negative sampling. Let C and W be the context and word embedding matrices respectively, where row c_t and w_t denote the context and word vector for token t.

Target word w	Context Word c	label y
jam	peach	+
jam	desk	-
jam	study	-
study	desk	+
study	jam	-

Table 1: Training Data

After training on mutiple epochs on the above training data, which row vectors in C and W might be different from their initial values? Explain why. (Write your answer in terms of c_t and w_t, e.g. use c_{jam} to represent the context vector for "jam".)

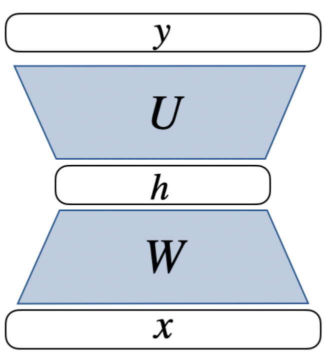
4 Feedforward Neural Networks (20 points)

1. (4 points) In one sentence, what is the purpose of a **softmax layer** in a neural network?

2. (4 points) In one sentence, what is the purpose of the **activation function** in a neural network?

3. Consider constructing a feedforward neural network (FFNN) for the task of **trigram language modeling**.

   (a) (4 points) Provide a diagram of your network. Your network should have the same level of detail as the example below. (This may not be the network you want for the trigram language modeling task!)



(b) (4 points) What is purpose of the input layer of your network? What is its dimension (i.e., size)? Explain your answer.

(c) (4 points) What is purpose of the output layer of your network? What is its dimension (i.e., size)? Explain your answer.